



# dxagent

Agentic diagnostic reasoning for dyspnoea and chest pain

*Validation meeting — what's built, what's honest, what's still open*

## THE PROBLEM

# Why a differential needs more than an LLM prompt

### No provenance

A single-prompt diagnosis carries no verifiable source for any claim it makes.

### No calibration

A rank is not a confidence — the model can be as sure about a guess as about a certainty.

### Always answers

It responds even when the evidence genuinely cannot decide — the failure mode that matters most in medicine.

*A system that doesn't know when to stay silent is more dangerous than one that's sometimes wrong.*

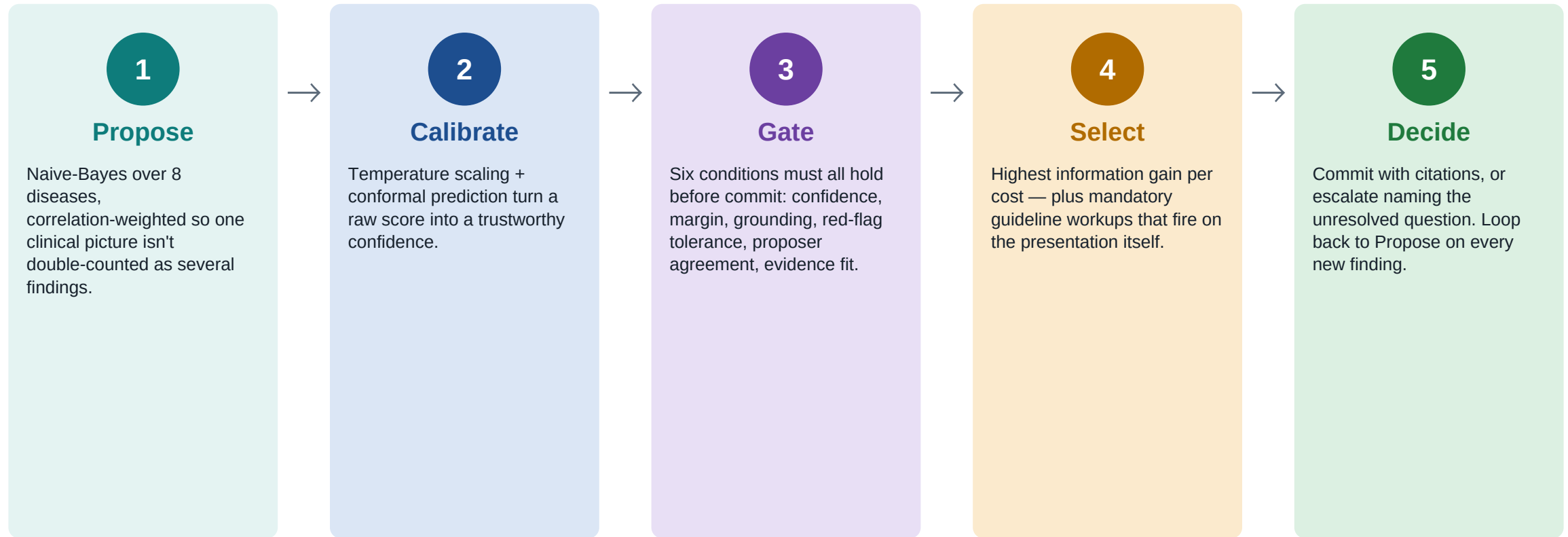
## WHAT IT DOES

# From patient findings to a cited, calibrated differential

- 1 Eight modelled diagnoses across dyspnoea and chest pain — three of them time-critical
- 2 One useful question or test at a time, chosen by expected information gain per unit cost
- 3 The belief over all eight diagnoses updates after every single answer
- 4 Every hypothesis carries citations back to a real, retrieved source — never authored by a model
- 5 Either commits with cited evidence, or escalates naming exactly what's unresolved

*Not a chatbot — there is no free-text interface. A decision loop that redraws itself after every answer.*

# The reasoning loop — five steps, run every turn



## WHAT'S BUILT

# Every number below is tested code, not a plan

159

tests in the suite

10/10

fixture cases correct

37%

likelihoods sourced

8

real PMC cases evaluated

### Bayesian reasoner + LangGraph engine

Two independent implementations of the same loop, proven equivalent by test

### RAG retrieval

BGE-M3 + Qdrant over 46 passages from 4 decision rules and the Merck Manual

### Clinical ontology

8 causes, 27 findings, 84 labelled edges — derived from the KB, consistent by construction

### Guideline workups

Wells / PERC / CURB-65 / HEART, plus mandatory + confirmatory tests that bypass a wrong posterior

### PHI screening, two layers

Regex for structured identifiers, spaCy NER for names inside free narrative text

### Live interactive consult

scripts/consult.py — a real person plays the patient, turn by turn

# Honestly — every number tagged by where it came from

## 97 of 153 likelihoods invented

Both bulk routes proven exhausted, not untried — DDXPlus by running its own extractor, Merck by two full passes over all 8 chapters.

Deleting them drops fixture accuracy 10/10 to 1/10. Individually insensitive, collectively load-bearing.

## 8 of 8 disease priors sourced

Derived from StatPearls presentation-conditional aetiology — the right quantity, found only after an earlier search targeted the wrong one.

Bands span a real 20-fold disagreement between sources on ACS.

| Source                         | Entries | What it is                                                                          |
|--------------------------------|---------|-------------------------------------------------------------------------------------|
| DDXPlus                        | 13      | Frequencies counted in ~1M simulated patients — a generating model, not real people |
| Merck Manual 19e               | 14      | Narrative phrases converted through a fixed, pre-committed rubric                   |
| Published cohorts / StatPearls | 29      | Miniati 2012 (PE), Zègre-Hemsey 2018 (ACS), StatPearls (pericarditis)               |

Most recent addition: pericarditis' ECG-change likelihood, invented at 0.60, sourced from StatPearls at 0.50 — lower than the invented value, kept anyway.

REAL PATIENTS

8 cases: 3 correct, 0 wrong, the rest escalated

| Case                                                   | True diagnosis               | Result                                                          |
|--------------------------------------------------------|------------------------------|-----------------------------------------------------------------|
| 78-year-old man, seven days of progressive shortnes... | Pulmonary embolism           | Committed — correct                                             |
| 64-year-old woman, 12 hours of chest pain and short... | Pericarditis                 | Escalated — ranked 3; top-1 confidence 28% below threshold 65%  |
| 32-year-old man, burning pleuritic chest pain radia... | Acute coronary syndrome      | Escalated — ranked 4; Pulmonary embolism exclusion could not... |
| 19-year-old woman, sudden trouble breathing and che... | Panic attack                 | Escalated — ranked 4; Pulmonary embolism exclusion could not... |
| 23-year-old man, one day of right-sided pleuritic c... | Community-acquired pneumonia | Committed — correct                                             |
| 45-year-old woman, two weeks of worsening shortness... | Acute pulmonary oedema       | Committed — correct                                             |
| 56-year-old man, 20 days of worsening dyspnoea wors... | COPD exacerbation            | Escalated — ranked 1; Pulmonary embolism exclusion could not... |
| woman in her mid-30s, two days of shortness of brea... | Asthma exacerbation          | Escalated — ranked 5; Pulmonary embolism exclusion could not... |

0 wrong commits — that is the number to look at.

Calibrated abstention is the claim, so the count that matters is not how many it answered but how many it answered wrongly. Every escalation names what it sought and could not get — several say the source report never recorded a D-dimer. n is small and the cases were hand-picked for clarity, so this is a few real data points rather than an accuracy result, and no threshold was tuned to make any of them pass.

# Found, closed generally, and traced honestly

## The gap

MandatoryWorkup enforced ordering a D-dimer once PE was a live concern, but never escalated a positive result to the confirmatory CTPA — a real gap for any future patient, not just this case.

Fixed with a new ConfirmatoryWorkup rule, cited from the same Wells literature already in the codebase. All 159 tests still pass; fixture accuracy unaffected.

## Why it still didn't flip the case

The rule-out mechanism scores every candidate as  $\text{discrimination} \div (\text{cost} + 1.5)$ . At the decisive turn it ranked a weak, cheap exam finding (JVP:  $0.27 \div 2.5 = 0.11$ ) above the far more decisive but expensive CTPA ( $\approx 0.9 \div 21.5 = 0.04$ ).

That scoring formula is shared with the ordinary selector and has a documented prior regression — not touched tonight, reported as an open structural finding for the roadmap instead of patched blind.



# Architectural properties, checked by tests — not a bolt-on policy

## No model authors a citation

Every citation is built by string-formatting on a passage that was actually retrieved — confirmed by code inspection, not assumed.

## Two independent loop implementations

A hand-written reference loop and a LangGraph state machine implement the same reasoning, proven equivalent by test.

## PHI screening, two layers

Regex catches structured identifiers; a spaCy NER pass reads the narrative and catches a name the regex is documented to miss.

## Six-condition abstention gate

Confidence, margin, grounding, red-flag tolerance, proposer disagreement, evidence-fit all have to hold before commit.

## WHAT'S STILL OPEN

# Named, not hidden

### The mandated model hasn't run live

No API credit all project. The LLM layer's been swappable since week 1 — the default now points at the current Opus-tier model as the honest equivalent, unverified live.

### AgentClinic

Named in the mandated stack. Genuinely blocked, not untried — needs an OpenAI/Replicate key plus forking its code to accept this agent.

### The constraint behind almost everything else

The brief's own clinical-partnership table calls for weekly meetings with a collaborating physician, feedback “part of your evaluation.” No clinician was available. That single fact is why every real patient came from a published case report rather than a clinician-reviewed one, and why the knowledge base still leans on DDXPlus and Merck rather than expert-elicited frequencies.

# Thank you.

Questions?